AI Support for Medical Diagnosis of Breast Cancer

Team 1

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Problem Statement

- **Breast Cancer Incidence**

  0.7% of all Mammographies are positive

- **Increased Breast Cancer**

  in 1975, compared to 2020, 9% to 12%

- **Global Deaths**

  according to WHO 685,000 global deaths of breast cancer occurred

- **AI outperforms doctors diagnosis of Breast cancer**

  There was a reduction of 1.2% in false positives, when a mammographie is incorrectly diagnosed

Stakeholders

Individuals

- Women > 35 years
- Men
- Doctors and Nurses
- Engineers

Groups

- WHO
- Government
- Insurance
- Pharmaceutical Company
- Hospital
- Minorities
- Engineering Company

Challenges

Data Collection

Is there an balanced enough data set to train a proper AI

Inequality

- Financial
- Education
- Social
- Geographical (Rural, City)

Transparency

Patients shall be aware of the AI process, so that they more easily consent for their data usage

Mistrust

- Patients
- Government
- Hospital Staff

AI Governance Framework

Pitch

Team 1

Educational

EDUCATION

- Training
- Awareness
- Promotion
- Prevention

Politcal

Social

STANDARDIZATION

- Expert groups
- Prevention
- Funding & Affordable AI
- Improvement

Pitch

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Ethics Canvas

Project Title: Clinical Diagnosis Driven by AI

Date: 21/06/22

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Individuals affected

Identify the types or categories of individuals affected by the product or service, such as men/women, user/non- user, age-category, etc.

- Women (>35)
- Men (indication)
- Younger women (indication)
- Doctors
- Nurses
- ~~Specialists~~ Hospital staff
- Engineers

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Behaviour

- not trained nurses
- bad scans
- no money (rural hospital)
- physical traits
- ignorant people

Relations

Discuss problematic differences in individual behaviour such as differences in habits, time-schedules, choice of activities, etc

- time savings
- accessible

What can we do?

Select the four most important Ethical impacts you discussed. Identify ways of solving these Impacts by changing your project's product/service design, organisation.Or by providing recommendations for its use or spelling out more clearly to users the values driving the design

Responsibility
↳ AI as support (multiple, expert group)

Standardization
→ globally expand
↳ rural areas
→ money issue

- Trusting unbiased AI
↳ trusting technology older people (prefer traditional)
↓
AI as support, change perception
→ usual resources

Worldviews

Discuss how the general perception of somebody's role in society can be affected by the project.

insurance ⟷ hospital
↓ government
↓ ethics

- money
- responsibility
- legal

e.g. employers and unions

- AI not explainable
↳ doctor only training
- training for hospital staff
- consent to use data, privacy
- dataset, first one

Groups affected

Identify the collectives or communities, e.g. groups or organisations, that can be affected by your product or service, such as environmental and religious groups, unions, professional bodies, competing companies and government agencies, considering any interest they might have in the effects of the product or service.

- Hospital
- pharmaceutic company
↳ screen
- insurance
- government
- WHO
- Engineering company
- family
- Minorities

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- privacy
↳ ask people for consent

Proc

Discus
service
error, f
breach

- Data losses
- Doctor blind-trusts
- not sufficient infrastructure
- long-term:
→ doctors ~~don't~~ unlearn how to detect

adapted from Alex Osterwalder's Business Model Canvas. The Business Model Canvas is designed by: Business Model Foundry AG. This work is
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